Lesson 1 - What is AI and Machine Learning?

READ ME FIRST (theory). Then open the practice notebook.

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Lesson 1 - What is AI and Machine Learning?

This is the **theory**. Read it slowly. There is almost no code in this lesson - it is about the **ideas and words** you must know. Then open `practice.ipynb`.

Key words in this lesson

Word	Simple meaning
AI	making computers do smart things
ML	a computer that learns from data

Word	Simple meaning
DL	ML that uses big “neural networks”
label	the correct answer we already know
feature	one piece of input information (a column)
model	the thing that learns the pattern and predicts

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1. AI, ML, DL - three circles

- **AI (Artificial Intelligence)** = the big idea: make a computer act smart.
- **ML (Machine Learning)** = a part of AI: the computer learns a pattern from data, instead of us writing every rule.
- **DL (Deep Learning)** = a part of ML: it uses big “neural networks” for hard things like images and speech.

Picture three circles: AI is the biggest, ML is inside AI, DL is inside ML.

Why it matters: people mix these words. Now you know AI is the biggest and DL is the smallest.

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2. What does “learn from data” mean?

Old way: a human writes every rule (“if word = ‘free’ then spam”). ML way: we show the computer many examples **with the answers**, and it finds the pattern by itself.

- **Feature** = an input column (for example: tenure, monthly charge).
- **Label** = the answer we already know (for example: churned = Yes/No).

Why: real life has too many rules to write by hand. Learning from examples scales.

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3. The 3 types of Machine Learning

Type	Do we have labels?	Goal	Example
Supervised	Yes (answers given)	Learn from answers, predict new ones	Past loans labeled “paid”/“not paid” -> predict new loans
Unsupervised	No	Find groups or structure	Group 1 million customers into similar types
Reinforcement	No, but there are rewards	Learn by trying + reward/penalty	A program learns a game by getting points

Common mistake: thinking everything is supervised. If there are **no labels**, it is unsupervised.

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4. Two kinds of supervised tasks

- **Regression** = predict a **number** (price, sales, temperature).
- **Classification** = predict a **category** (spam / not spam, churn Yes / No).

Quick test: "How much?" -> regression. "Which group?" -> classification.

Common mistake: calling churn (Yes/No) a regression. It is a **category**, so it is classification.

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5. The ML workflow (the steps we follow)

1. **Understand the problem** - what do we want, and why?
2. **Get the data.**
3. **Clean and prepare** the data (Modules 2-3).
4. **Train a model** (Module 4).
5. **Check how good it is** (Module 4).
6. **Use it** on new data.

Why: if you skip step 1, you can build a perfect model for the **wrong** problem.

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6. A model is never 100% sure

A model gives a **probability**, not a certain answer. A weather model says "80% rain", not "it will rain for sure". The future has uncertainty.

Why: this is normal and fine. We measure how often the model is right, and try to improve it.

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7. Fairness and bias (short but important)

If the data is unfair, the model is unfair. Example: a model trained mostly on data from men may work badly for women. This is called **bias**.

Why: we must check who the data represents, so the model is fair to everyone.

Quick summary

- AI > ML > DL (size).

- ML learns from data: **features** (inputs) + **labels** (answers).
- 3 types: **supervised** (labels), **unsupervised** (no labels), **reinforcement** (rewards).
- Supervised tasks: **regression** (number) or **classification** (category).
- Always understand the problem first. Models give probabilities. Watch for bias.

Now do this

1. Open `practice.ipynb`.
2. Do the scenario exercises (decide the ML type).
3. Run the small Python warm-up so you are ready for Lesson 2.